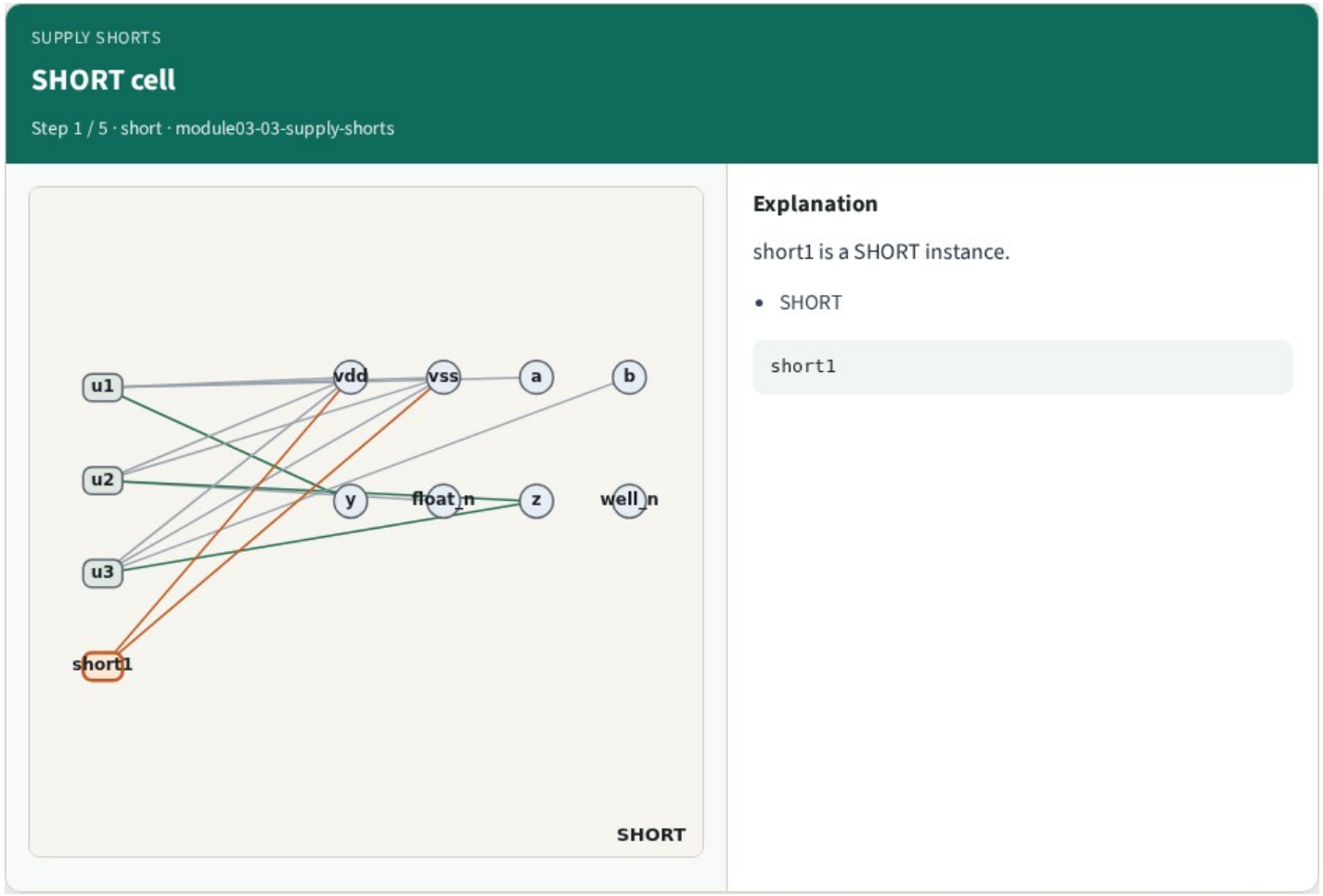
Why shorts

A hard short between vdd and vss is catastrophic ERC

The idea

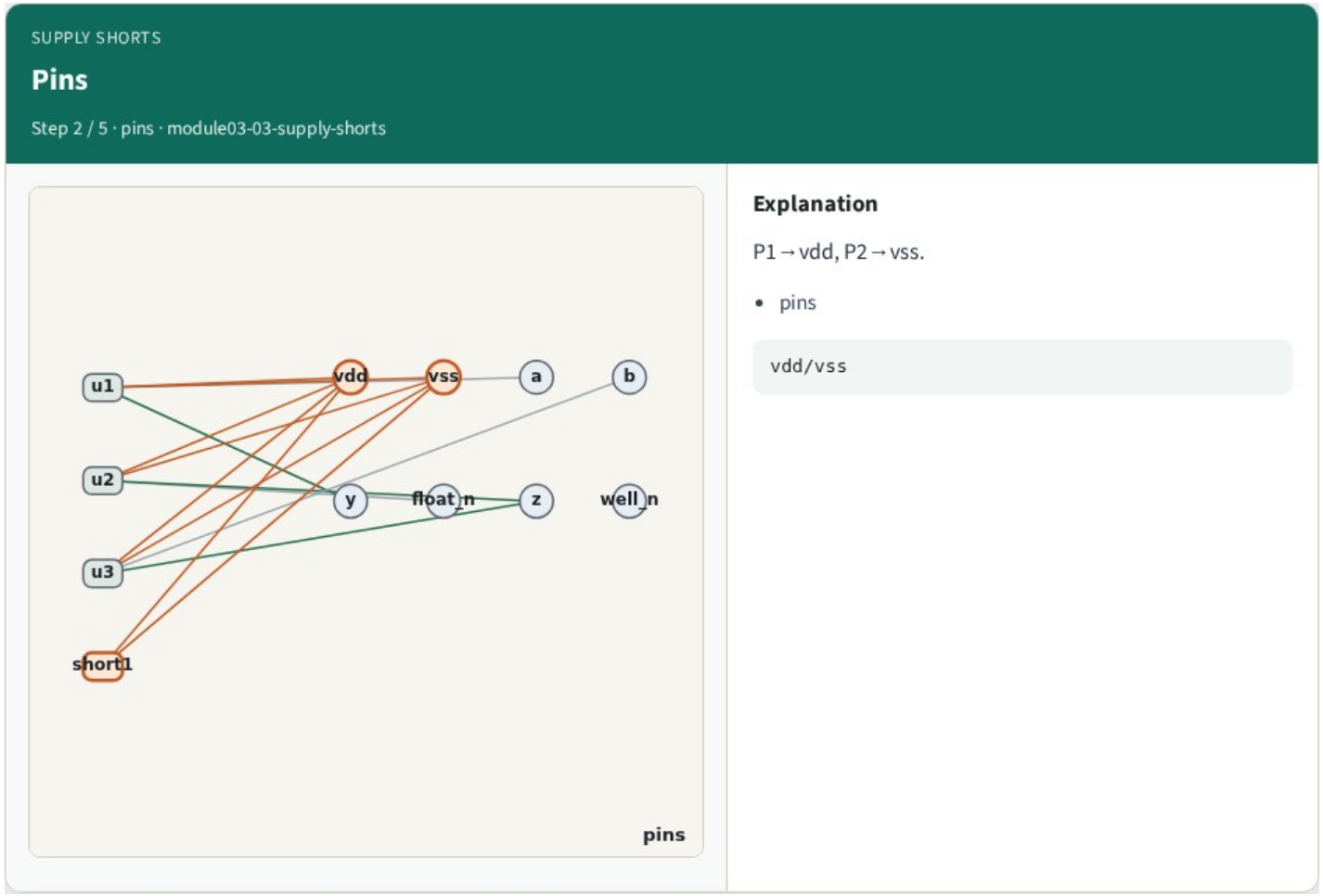
- GOLDEN: supply_short_count=1 via short1

SHORT cell



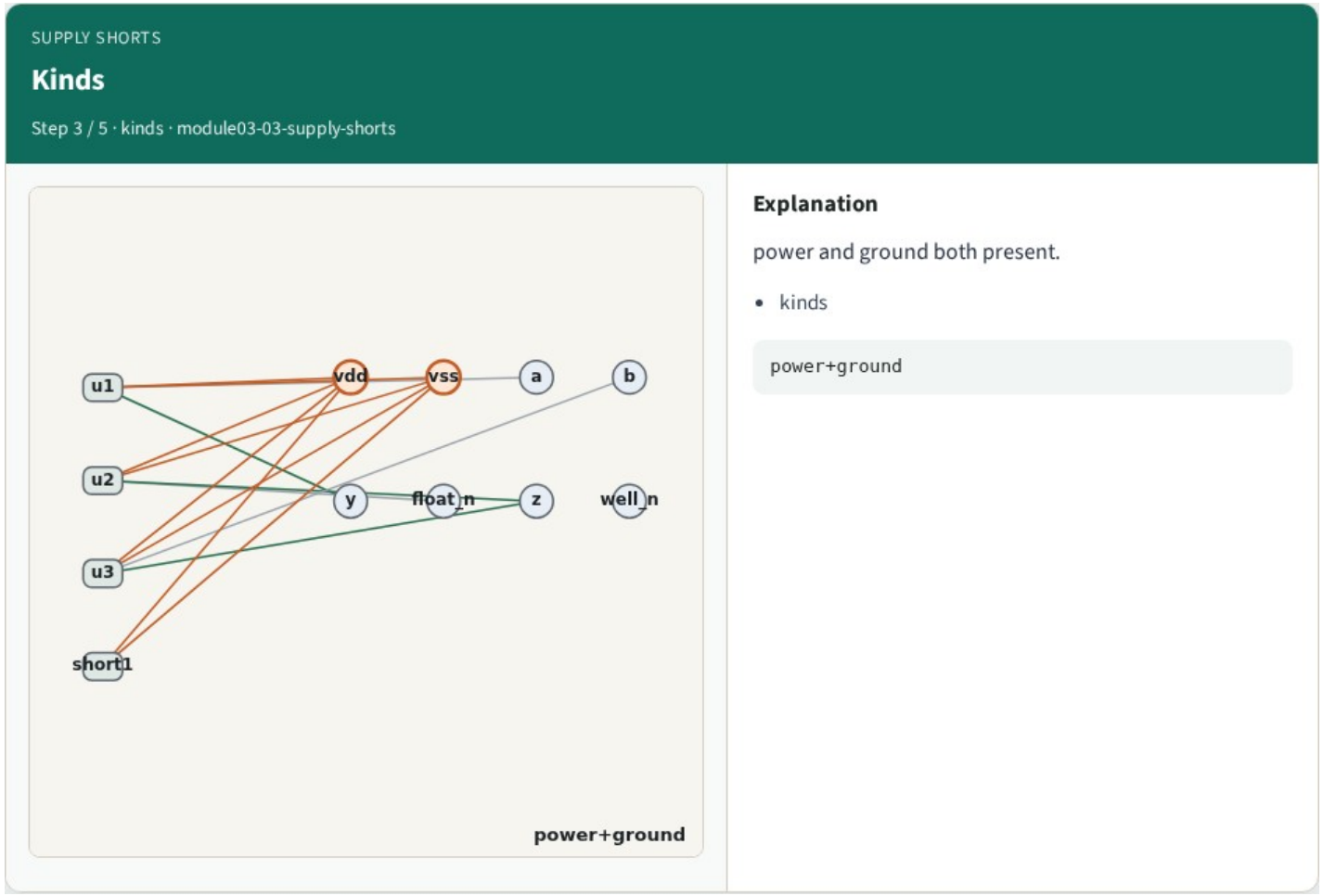
SHORT cell

Pins


SUPPLY SHORTS
Pins
Step 2 / 5 · pins · module03-03-supply-shorts
u1
u2
u3
short1
vdd
vss
a
b
y
float_n
z
well_n
pins
Explanation
P1 → vdd, P2 → vss.
pins
vdd/vss

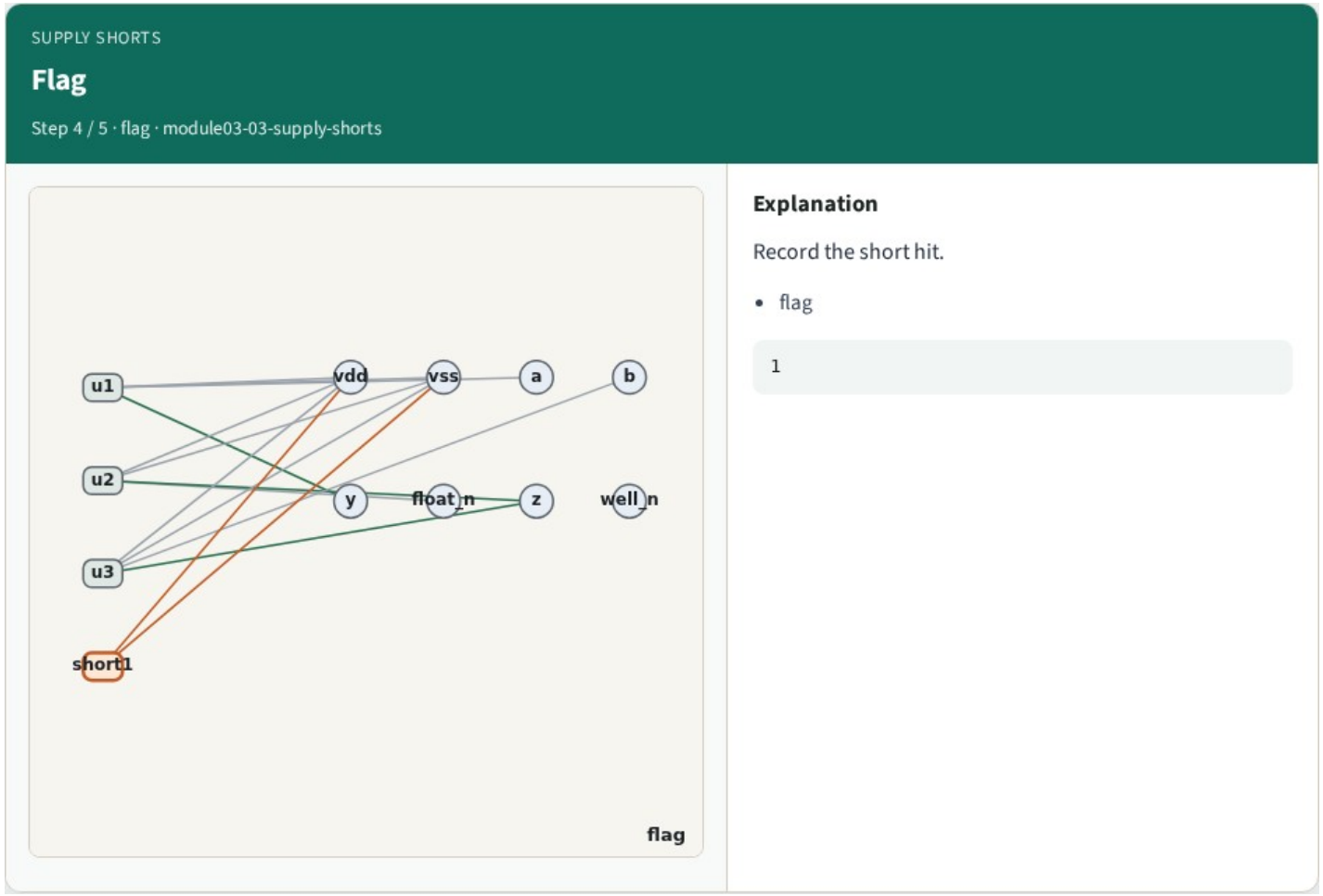
Pins

Kinds


SUPPLY SHORTS
Kinds
Step 3 / 5 · kinds · module03-03-supply-shorts
u1
u2
u3
short1
vdd
vss
a
b
y
float_n
z
well_n
power+ground
Explanation
power and ground both present.
kinds
power+ground

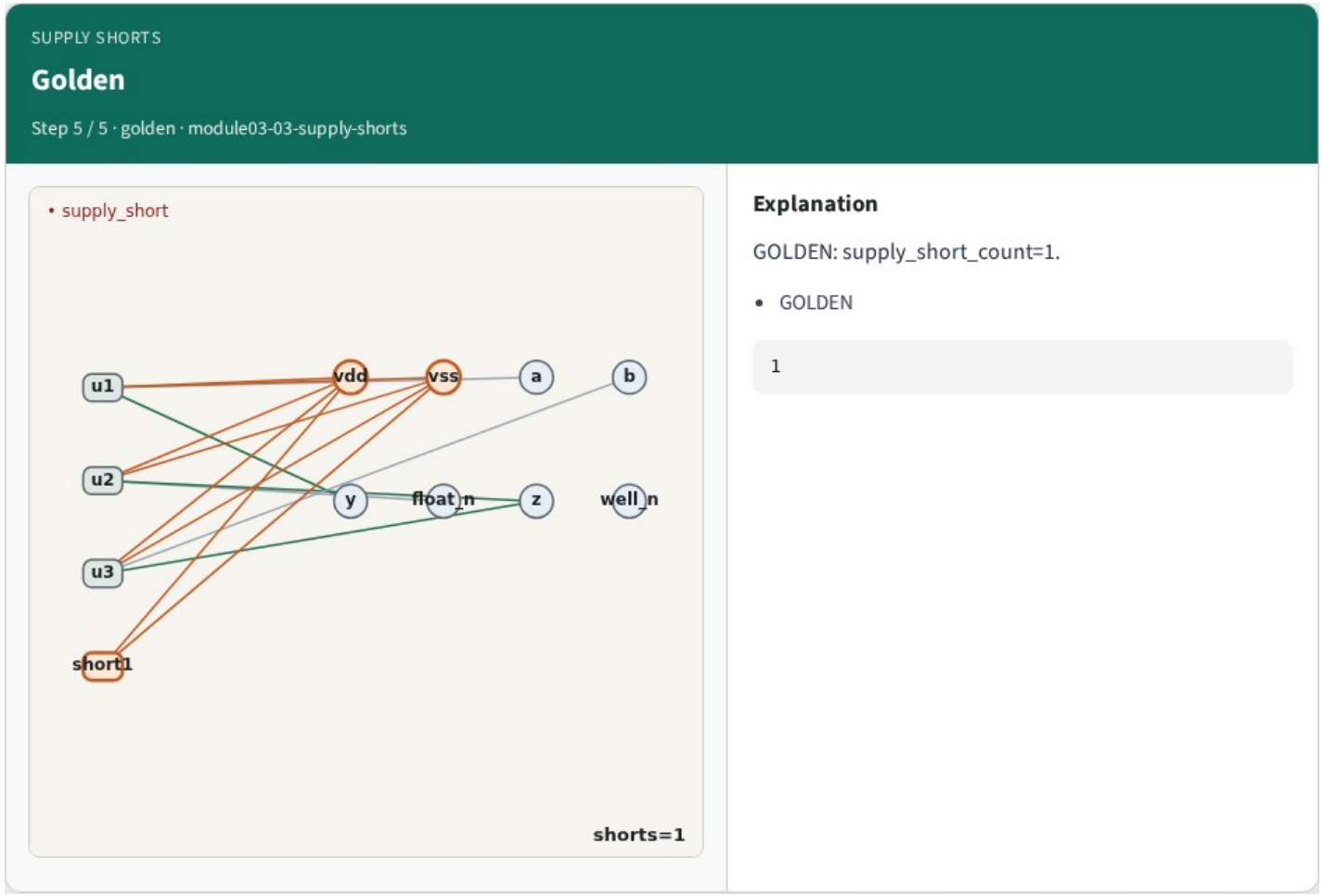
Kinds

Flag


SUPPLY SHORTS
Flag
Step 4 / 5 · flag · module03-03-supply-shorts
u1
u2
u3
short1
vdd
vss
a
b
y
float_n
z
well_n
flag
Explanation
Record the short hit.
flag
1

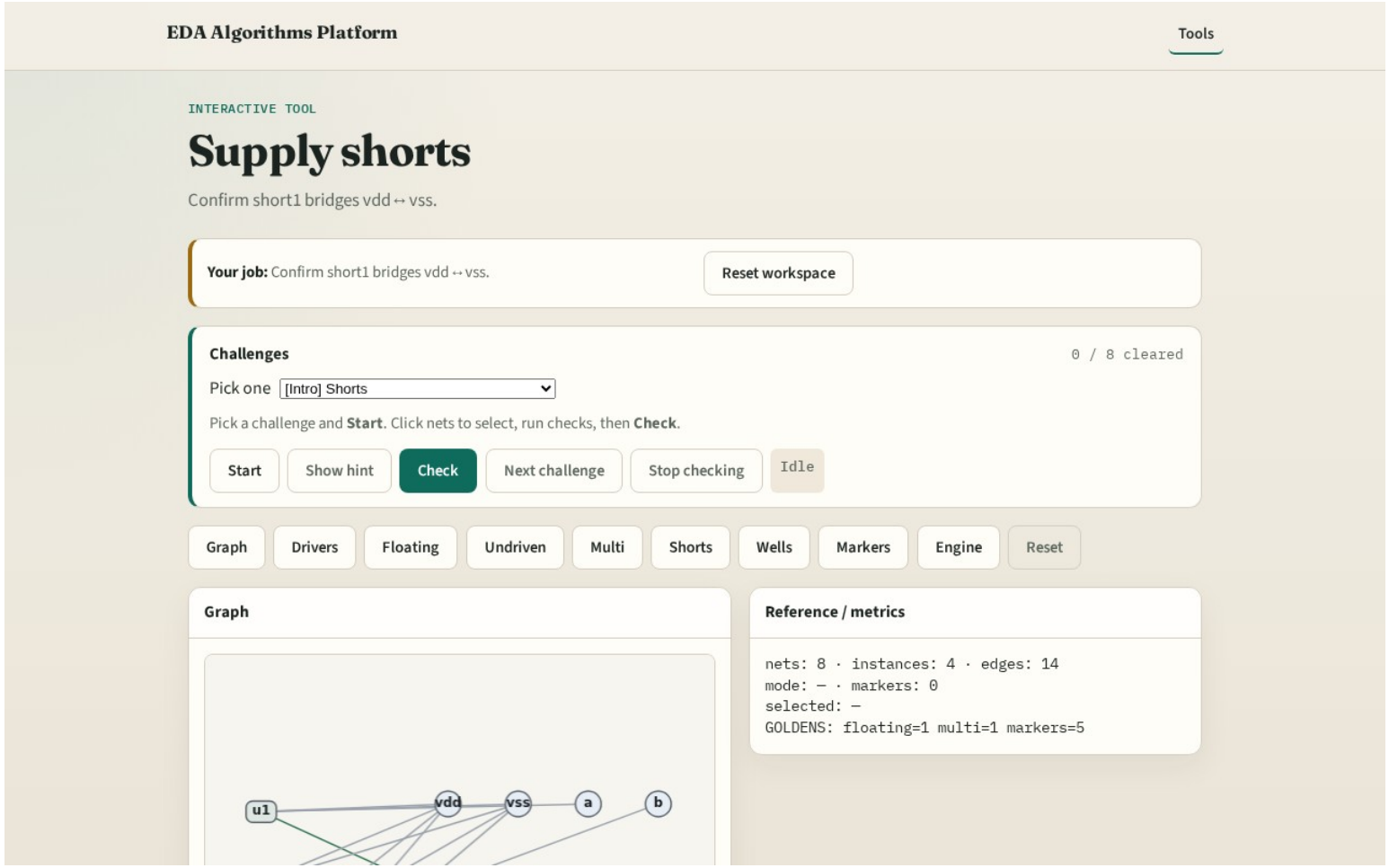
Flag

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call find_supply_shorts(); assert count==1

Pitfalls

- Treating normal INV VDD/VSS as a short
- Missing SHORT cell type

Your turn

- Next: well/substrate ties